3.3.3 Higher Order Symplectic Methods: The Suzuki-Yoshida Method

Yoshida [398] gives an elegant method for creating a symplectic scheme of arbitrarily high order, using Trotter's results for compositions of linear operators [370]. This work is related to methods suggested by Suzuki [354] in the context of Trotter factorization of quantum operators. Consider a scheme with order $2s$ (for $s \geq 1$) and where the evolution of the system under the method is given by $\exp(h\,\hat{\tilde{\mathcal{L}}}_h)$, where

$$\hat{\mathcal{L}}_h \stackrel{\text{def}}{=} \mathcal{L}_{\tilde{H}_h} = \mathcal{L}_H + h^{2s}\tilde{\mathcal{L}}_{[2s]} + h^{2s+2}\tilde{\mathcal{L}}_{[2s+2]} + h^{2s+4}\tilde{\mathcal{L}}_{[2s+4]} + \mathcal{O}(h^{2s+6})$$

is the method's characteristic operator (with corresponding shadow Hamiltonian $\tilde{H}_h$). The linear operators $\tilde{\mathcal{L}}_{[2i]}$ are compositions of Lie derivatives, and can be computed using the symmetric BCH formula. We now consider iterating this scheme three times, first using a step of $\tau_0 h$, followed by a step of $\tau_1 h$, and finally another step with stepsize $\tau_0 h$ (where $\tau_0 h + \tau_1 h + \tau_0 h = h$). The overall effect on the system is given by a product of exponentials, as

$$\exp\left(\tau_0 h\,\hat{\mathcal{L}}_{\tau_0 h}\right)\exp\left(\tau_1 h\,\hat{\mathcal{L}}_{\tau_1 h}\right)\exp\left(\tau_0 h\,\hat{\mathcal{L}}_{\tau_0 h}\right) = \exp\left(h\,Z_h\right),$$

where, using repeated application of the BCH formula, we find

$$\begin{aligned} Z_h &= 2\tau_0\,\hat{\mathcal{L}}_{\tau_0 h} + \tau_1\,\hat{\mathcal{L}}_{\tau_1 h} + \mathcal{O}(h^{2s+2}), \\ &= (2\tau_0 + \tau_1)\,\mathcal{L}_H + (2\tau_0^{2s+1} + \tau_1^{2s+1})h^{2s}\tilde{\mathcal{L}}_{[2s]} + \mathcal{O}(h^{2s+2}). \end{aligned}$$

We have free reign over constants τ_0 and τ_1 as long as $2\tau_0 + \tau_1 = 1$. Hence we have an opportunity to annihilate the perturbation operator at order h^{2s} by choosing $2\tau_0^{2s+1} + \tau_1^{2s+1} = 0$ as well. Solving simultaneously, there exists a unique real solution

$$\tau_0 = \frac{1}{2-\kappa}, \qquad \tau_1 = -\frac{\kappa}{2-\kappa}, \qquad \kappa^{2s+1} = 2,$$

giving us a scheme of order $2s+2$. We can then proceed recursively, as this new order $2s+2$ scheme can be composed similarly to wipe out successive higher order terms.

Though the fourth-order version of the scheme was given first by Forest and Ruth [139] (and discovered independently by Yoshida [398] and Candy and Rozmus [66]), we shall simply refer to these higher-order schemes as Yoshida methods, owing to the elegant derivation of schemes of arbitrary order.

Example 3.1 Consider using velocity Verlet as the base second-order method to build a Yoshida fourth-order method from. As the scheme is second-order, we have